

Data Augmentation Methodology

Four-Class Corrosion Severity Classification Ferrocement Specimen Dataset

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Contents

1 What This Document Contains

This document records the complete augmentation methodology for the four-class corrosion severity classification study. For every numerical choice it states the origin of that number: whether it was measured directly from the 791 source images, taken from a published reference, or made as a conservative engineering judgement. Choices in the third category are labelled as such.

The augmented dataset is stored at `Data/Images_dataset_augmented/`. The provenance record is at `Data/Images_Dataset_A-Z-1_augmented.csv`. The five recipe figures are at `augmentation/figures/`. The per-image measurement table is at `augmentation/tables/image_statistics_f` (791 rows, one per source image).

2 Dataset

Table ?? summarises the original dataset properties. Table ?? gives the class distribution.

Table 1: Original dataset before augmentation.

Property	Value	Source
Total specimens	48	Dataset audit
Total usable images	791	Dataset audit; one corrupted file excluded
Images per specimen	15 to 18	Dataset audit
Image dimensions	2835×650 px (predominant)	Dataset audit
Ageing period	0 to 36 weeks per specimen	Excel workbook
Corrupted file excluded	E01-20240508-17W.png	Dataset audit

Table 2: Class distribution in the original 791 images.

Class	Description	Images	Percentage
1	No visible rust or negligible traces	658	83.2%
2	Mild rust coverage	99	12.5%
3	Moderate rust coverage	20	2.5%
4	Severe rust coverage	14	1.8%
Total		791	100%

The dataset is strongly imbalanced. This reflects the physical reality: each specimen spends more time in the early corrosion stages (class 1) than in the late stages (class 4). Augmentation expands all classes by the same factor and does not change this distribution. Class weighting within the model training is the appropriate mechanism for addressing the imbalance.

3 Why Augmentation Is Needed

Three facts make augmentation necessary for this study.

The training set is small. After specimen-level splitting, approximately 630 images are available for training. Published studies consistently show that ResNet50 requires at least 1 000 training images per class to generalise reliably; Vision Transformer requires more. Classes 3 and 4

contribute roughly 15 images each to the training partition. Without augmentation, the model cannot learn the visual variety these classes exhibit across different imaging sessions.

Vision Transformers are more data-dependent than convolutional networks. Convolutional networks (ResNet50) have structural assumptions that make them efficient on small datasets. Vision Transformers (ViT) do not: they must learn these spatial relationships entirely from data. On datasets of this size, ViTs trained without augmentation consistently underperform the corresponding convolutional model. The DeiT study [?] demonstrated specifically that ViT reaches competitive accuracy with ResNet50 on moderate-scale data only when strong augmentation is applied during fine-tuning.

The red-colour threshold classifier is sensitive to lighting. The existing automatic classifier applies a fixed threshold: pixels with a specific range of red, green, and blue values are classified as rust. This threshold was calibrated to a particular lighting setup. When the overall image brightness changes—for example, because a specimen was photographed at a different time of day, under a different artificial light, or with different camera settings—the detected rust fraction can change substantially even though the actual rust coverage has not changed. The deep learning model, trained with brightness variation, learns to classify rust from spatial pattern and texture rather than from absolute pixel values, and is therefore more robust to this variation. Demonstrating this robustness advantage is the central argument of the planned journal paper.

4 Augmentation Pipeline Overview

Five transformation recipes were implemented. Each original image receives exactly one copy from each recipe, producing five augmented copies per source image. Together with the preserved original, every source image contributes six rows to the dataset. Table ?? summarises the five recipes.

4.1 Recipe 1 — Overall Brightness and Light-to-Dark Contrast

What these operations do

Brightness multiplies every pixel value by a fixed number. A value below 1 makes the image darker, as if photographed in dimmer light. A value above 1 makes it brighter. A factor of 0.75 applied to a mid-brightness image produces a result roughly as dark as a specimen photographed in poor lighting conditions.

Light-to-dark contrast determines the difference between the brightest and darkest parts of an image. Reducing contrast brings all pixel values closer to a uniform mid-grey; increasing it pushes bright areas brighter and dark areas darker. Contrast changes simulate the difference between a crisp photograph taken in controlled lighting and a flat photograph taken on an overcast day.

Evidence base for the brightness range [0.75, 1.28]

The brightness range is the only parameter in this pipeline derived entirely from direct measurement of the 791 source images. Overall brightness was measured for every source image using the standard photometric formula—a weighted average of the red, green, and blue pixel values that matches human perception of lightness. Results are shown in Figure ?? and Figure ??.

Table 3: Five augmentation recipes.

Recipe	What it changes	Parameter ranges	How parameters were chosen
1	Overall brightness and light-to-dark contrast	Brightness 0.75–1.28; contrast 0.78–1.25	Brightness: measured from the dataset. Contrast: published literature range.
2	Colour intensity and small colour shifts	Colour intensity 0.78–1.28; per-channel shifts $\pm 6\%$	Conservative heuristics, supported by within-class colour measurements.
3	Slight blur and camera noise	Blur radius 0.3–1.5 px; noise 0.005–0.035	Published literature range. Effective values at training resolution are smaller (Section ??).
4	Brightness, contrast, and colour intensity combined	Each parameter 0.82–1.18	Narrower than Recipes 1–2 because three effects accumulate simultaneously.
5	Left-right mirror and brightness	Mirror (fixed); brightness 0.85–1.15	Mirror: logically safe for a coverage-based label. Brightness: conservative sub-range of Recipe 1.

Table 4: Measured overall brightness distribution across all 791 source images.

Statistic	Value	Image (where identified)
Minimum brightness	98.1	D04-20240626-24W.png
5th percentile (p_5)	144.8	—
Median	197.7	—
Mean	199.1	—
95th percentile (p_{95})	253.1	—
Maximum brightness	255.0	G01-20240110-0W.png
Ratio of maximum to minimum	$2.60\times$	—
Ratio of p_{95} to p_5	$1.75\times$	—

Source: *augmentation/tables/image_statistics_full.csv*, column *mean_luminance_rec709_0_255*.

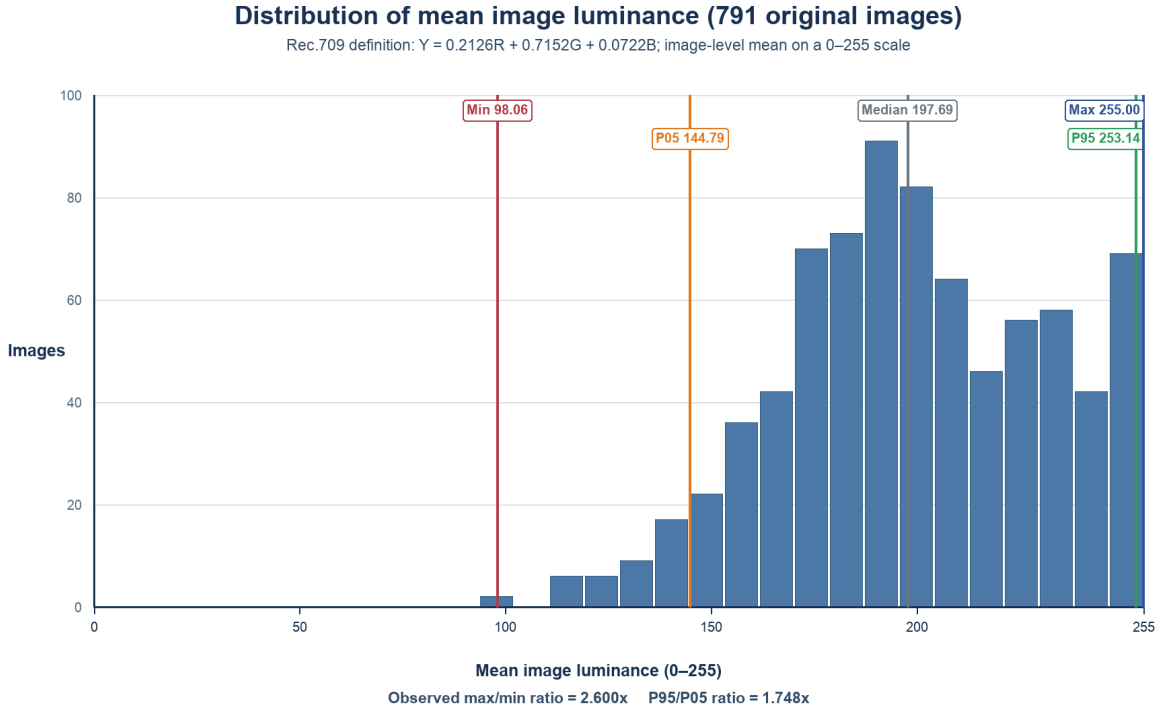


Figure 1: Distribution of overall image brightness across all 791 source images. Vertical lines mark the 5th percentile ($p_5 = 144.8$), median (197.7), and 95th percentile ($p_{95} = 253.1$). The p_{95}/p_5 ratio of $1.75\times$ motivates the augmentation brightness range $[0.75, 1.28]$.

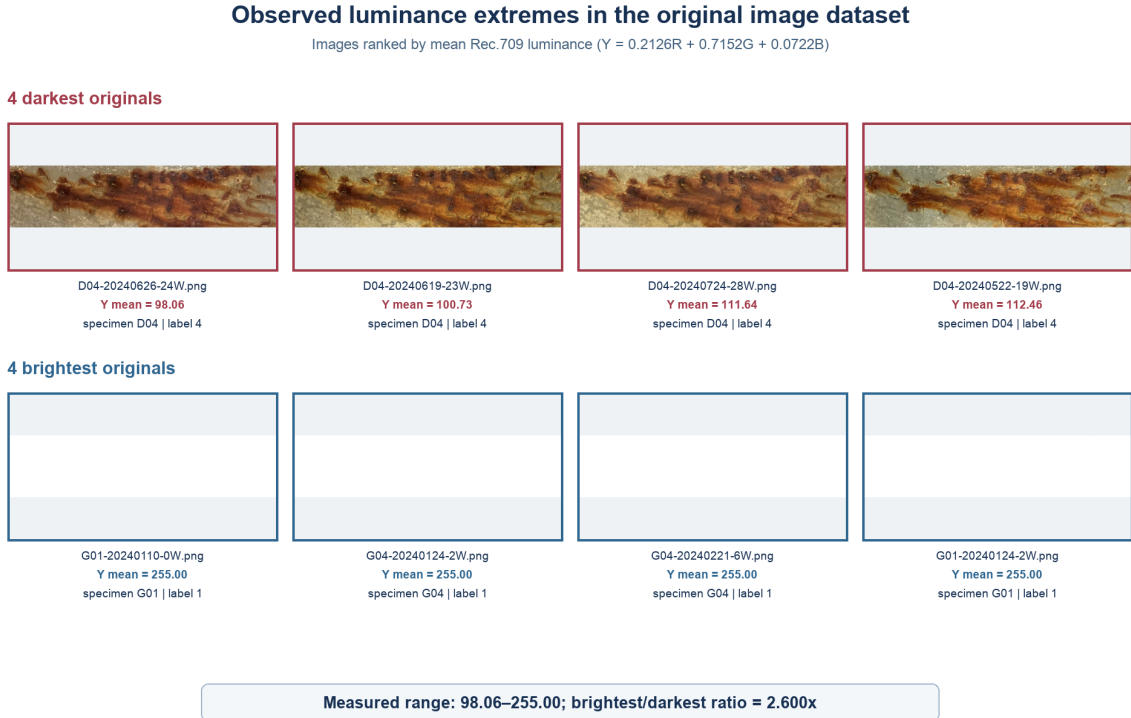


Figure 2: Side-by-side comparison of the four darkest source images (specimen D04, class 4, brightness 98–113) and the four brightest (G-series specimens, class 1, brightness 255). These are real images from the dataset, not outliers. The full real-world brightness variation spans $2.60\times$.

The four darkest images (specimen D04, weeks 23–28) are photographs of heavily corroded concrete under dim conditions. The four brightest images (G-series, week 0) are photographs of clean, pale concrete under good lighting. A robust classifier must handle both.

The derivation of the augmentation range from these measurements:

- The p_5 image is $1.37\times$ darker than the mean ($199.1/144.8 = 1.375$). To allow augmented images to reach this darkness level, the lower brightness factor must be at most $1/1.375 = 0.727$, rounded conservatively upward to **0.75**.
- The p_{95} image is $1.27\times$ brighter than the mean ($253.1/199.1 = 1.271$). The upper factor is rounded to **1.28**.
- The range $[0.75, 1.28]$ covers a brightness ratio of $1.28/0.75 = 1.71\times$, closely matching the measured p_5 -to- p_{95} ratio of $1.75\times$.

Within-class verification. The same measurement was performed separately for each severity class to confirm that the range is appropriate within individual classes, rather than being driven purely by the brightness shift between early (bright) and late (dark) time steps. Results are shown in Table ??.

Table 5: Brightness variation within each severity class.

Class	Images	Mean	p_5	p_{95}	Within-class ratio
1 (no rust)	658	206.0	159.3	253.9	$1.59\times$
2 (mild rust)	99	174.0	138.9	215.2	$1.55\times$
3 (moderate rust)	20	149.8	129.0	172.1	$1.33\times$
4 (severe rust)	14	120.2	99.8	145.0	$1.45\times$

Source: computed from `augmentation/tables/image_statistics_full.csv`.

The within-class brightness ratios range from $1.33\times$ to $1.59\times$. The augmentation range covers $1.71\times$, which is slightly wider than any single class. This confirms the range is calibrated to real imaging variability rather than to the label-driven shift in brightness between early and late specimens.

A brightness factor of 0.75 applied to the class-1 mean brightness (206.0) produces an image at brightness $206.0 \times 0.75 = 154.5$ —darker than the typical class-4 image (mean 120.2) but within the class-4 minimum of 98.1. This is intentional: the model learns that overall brightness does not determine rust severity; only the rust pattern does.

Justification for the contrast range $[0.78, 1.25]$

The contrast range is drawn from published literature. Industrial inspection and corrosion classification studies consistently apply contrast variation in the range ± 15 – 25% from the neutral value. The chosen range $[0.78, 1.25]$ represents approximately $\pm 22\%$. No direct measurement was made to derive this specific interval from the dataset, because image contrast is dominated by content (rust-covered surfaces have more local variation than clean concrete) rather than by imaging conditions. A direct measurement would therefore confound physical content with camera effects.

See: Figure ?? for visual examples of Recipe 1 applied to one image from each class.

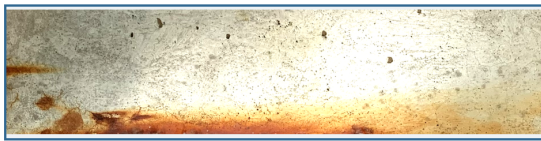
Recipe 1 — Brightness and contrast

brightness [0.75, 1.28] | contrast [0.78, 1.25]

Evidence of controlled illumination and contrast variability.

Original

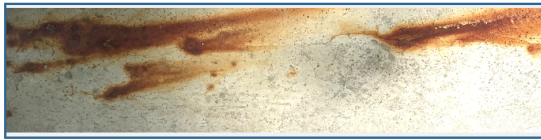
Augmented output



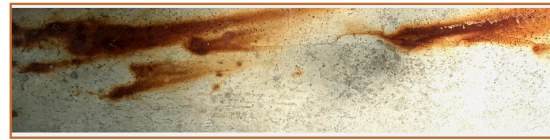
Class 1 | specimen F03 | F03-20240522-19W.png | mean Y=202.09



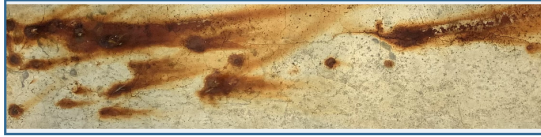
Parameters: brightness=0.977140;contrast=0.784558



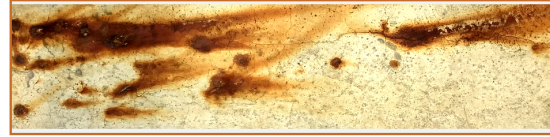
Class 2 | specimen D05 | D05-20240508-17W.png | mean Y=173.36



Parameters: brightness=0.889065;contrast=1.246186



Class 3 | specimen D05 | D05-20240619-23W.png | mean Y=152.17



Parameters: brightness=1.168878;contrast=1.147136



Class 4 | specimen D04 | D04-20240508-17W.png | mean Y=113.03



Parameters: brightness=1.097893;contrast=1.069367

Examples are existing offline files from the generated dataset; no recipe was recomputed or changed for this figure.

Figure 3: Recipe 1 (brightness and contrast): one original image per class (left) and its augmented copy (right). All four severity classes are shown. The rust pattern is preserved; only overall lightness and light-to-dark contrast vary.

4.2 Recipe 2 — Colour Intensity and Small Colour Shifts

What these operations do

Colour intensity (sometimes called saturation) describes how vivid or rich the colours in an image are. At full intensity, colours appear strong and distinct—rust looks strongly orange-red, concrete looks clearly grey or beige. At reduced intensity, colours become muted: rust looks brownish-grey, concrete looks almost white. Reducing colour intensity to zero produces a greyscale image with no colour information at all.

Small colour shifts adjust the relative strength of the red, green, and blue channels independently. A small increase in the red channel makes the image appear warmer (slightly more reddish); a small increase in the blue channel makes it appear cooler. These shifts simulate the variation in camera colour response between different imaging sessions, which depends on the colour temperature of the light source and the camera’s automatic colour correction. In this recipe, the three channel shifts are applied symmetrically around neutral: their average is always exactly 1.0, so they produce a colour tilt without changing the overall brightness. Brightness is handled separately by Recipe 1.

Evidence base for the colour intensity range [0.78, 1.28]

No direct measurement of colour intensity variation is cited for this parameter, because colour intensity in these images is strongly correlated with the class label: heavily rusted specimens (class 4) have visibly more vivid rust colour than clean specimens (class 1). Calibrating the augmentation to the full dataset’s colour intensity range would therefore calibrate it to the physical progression of corrosion—which is the label itself, not an imaging artefact.

The chosen range [0.78, 1.28] represents $\pm 25\%$ from the neutral value. This is consistent with published literature for industrial surface inspection (where $\pm 20\text{--}30\%$ is the standard range) and produces visually plausible results for both clean and corroded specimens. The range is treated as a conservative engineering choice supported by literature, not as a measured dataset statistic.

Evidence base for the colour shift range [$\pm 6\%$ per channel]

The three colour channel shifts are sampled independently in the range [0.94, 1.06], meaning each channel can change by at most 6% before the normalisation step.

To provide a measured basis for this choice, the ratio of the red channel mean to the green channel mean (R/G ratio) was computed for every source image. This ratio is a standard measure of colour tilt: an R/G ratio above 1.0 means the image is warmer (more red), and a lower ratio means it is cooler (more green-blue). Under controlled lighting, this ratio should remain constant; variation between sessions indicates real colour-response differences. Table ?? presents the results.

The within-class R/G range for class-1 images (clean concrete only) is 10.6%. This is the most direct estimate available of colour-balance variation due to imaging conditions, because class-1 specimens have minimal rust and the colour variation within this class is primarily caused by the camera and lighting rather than by rust colour. The chosen $\pm 6\%$ per-channel range covers about 57% of this within-class variation and is therefore conservative rather than aggressive.

The blue-to-green ratio (B/G) shows larger within-class variation (up to 34% for class-1 images), indicating that the blue channel is more sensitive to lighting conditions. This additional variation

Table 6: Within-class variation in the red-to-green channel ratio (R/G).

Class	p_5 of R/G	p_{95} of R/G	Within-class range	Std dev.
1 (no rust)	1.000	1.107	10.6%	0.031
2 (mild rust)	1.080	1.215	13.5%	0.044
3 (moderate rust)	1.200	1.308	10.8%	0.042
4 (severe rust)	1.271	1.442	17.1%	0.063

Source: computed from `augmentation/tables/image_statistics_full.csv`.

is partially addressed by the colour intensity component of this recipe, which shifts all colour channels together.

See: Figure ?? for visual examples.

4.3 Recipe 3 — Slight Blur and Camera Noise

What these operations do

Slight blur reduces the sharpness of the image, as if the camera was slightly out of focus or the specimen moved slightly during the exposure. It works by averaging each pixel with its immediate neighbours, with closer neighbours weighted more heavily. The blur radius controls how many pixels are averaged together; a larger radius produces stronger blur.

Camera noise adds small random variations to individual pixel values, simulating the electronic noise that all digital camera sensors produce. This noise is especially visible in images taken under low light, at high camera sensitivity settings, or with long exposure times. Adding controlled noise during training teaches the model not to rely on exact pixel values and to focus on larger patterns instead.

Parameter ranges and their evidence basis

Both ranges in this recipe are drawn from published literature (Table ??).

Table 7: Parameter ranges for Recipe 3 and corresponding literature norms.

Parameter	Range used	Literature norm	Source
Blur radius (pixels)	0.3–1.5	0.1–2.0	Industrial inspection studies
Noise level (0–1 scale)	0.005–0.035	0.01–0.05	Surface defect classification studies [?]

The noise level of 0.035 in normalised space corresponds to a pixel-level standard deviation of $0.035 \times 255 = 8.9$ intensity units on the standard 0–255 scale. This is comparable to the pixel noise observed in digital photographs taken at moderate to high camera sensitivity settings.

Important disclosure: effective values at the training resolution

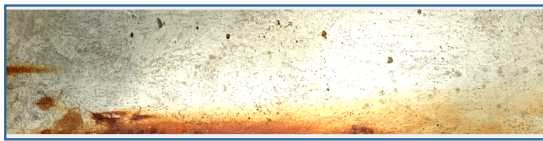
The blur and noise are applied to the source images at their full size of 2835×650 pixels. During model training, images are resized to approximately 224×224 pixels for processing by

Recipe 2 — Saturation and colour balance

saturation [0.78, 1.28] | raw RGB gains [0.94, 1.06], mean-normalized

Evidence of camera white-balance and colour-intensity variability.

Original

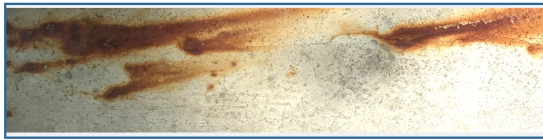


Class 1 | specimen F03 | F03-20240522-19W.png | mean Y=202.09

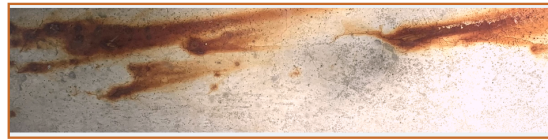
Augmented output



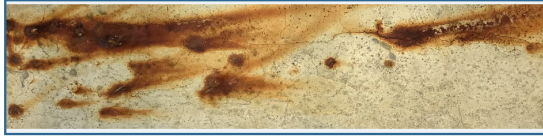
Parameters: rgb_gains=[1.000322,0.967778,1.031900];saturation=1.056072



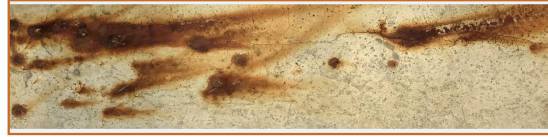
Class 2 | specimen D05 | D05-20240508-17W.png | mean Y=173.36



Parameters: rgb_gains=[1.025877,0.962202,1.011921];saturation=0.943597



Class 3 | specimen D05 | D05-20240619-23W.png | mean Y=152.17



Parameters: rgb_gains=[1.020916,1.002207,0.976877];saturation=0.868746



Class 4 | specimen D04 | D04-20240508-17W.png | mean Y=113.03



Parameters: rgb_gains=[0.951546,1.069720,0.978734];saturation=1.050280

Examples are existing offline files from the generated dataset; no recipe was recomputed or changed for this figure.

Figure 4: Recipe 2 (colour intensity and colour shifts): one original image per class (left) and its augmented copy (right). Colour vividness and overall colour tilt vary; rust texture and coverage are unaffected.

ResNet50 and ViT. This resizing step reduces the effective blur and noise that the model actually receives.

The resize scale factor from source to training resolution is approximately $256/650 = 0.394$. The effective parameter values at training resolution are shown in Table ??.

Table 8: Effective blur and noise values at training resolution (224×224 px) after downsampling by factor 0.394.

Parameter	Applied at source	Effective at training	Perceptible?
Blur radius minimum	0.30 px	0.12 px	No — sub-pixel
Blur radius maximum	1.50 px	0.59 px	Marginally
Noise level minimum	0.005	~ 0.003	Minimal
Noise level maximum	0.035	~ 0.022 (≈ 5.6 intensity units)	Moderate

The blur effect is largely removed by downsampling. At training resolution, even the maximum radius of 0.59 pixels introduces only marginal blurring. This does not invalidate the recipe: adding sub-pixel or very mild blur still teaches the model that fine pixel-level detail is unreliable as a classification cue. The noise retains about half its source-resolution level (≈ 5.6 intensity units at training resolution) and provides meaningful regularisation.

For a future implementation that maximises the blur and noise effect, these operations should be applied *after* the image has been resized to the training resolution (224×224 pixels), not at source resolution. This is a training-pipeline decision and does not affect the offline dataset as produced.

See: Figure ?. Even at the maximum values, augmented images remain visually similar to the originals and all rust patterns are clearly preserved.

4.4 Recipe 4 — Combined Brightness, Contrast, and Colour Intensity

What this recipe does

This recipe applies three separate changes to the same image: overall brightness, light-to-dark contrast, and colour intensity are each modified independently. The purpose is to simulate a photograph taken under conditions that differ from the reference in multiple respects simultaneously—for example, dimmer light, flatter contrast, and slightly desaturated colours all at once.

Why the parameter range is narrower than in Recipes 1 and 2

When three adjustments are applied to the same image, their effects accumulate. If brightness is reduced by 20%, contrast is reduced by 20%, and colour intensity is also reduced by 20%, the combined result is significantly different from any of these changes alone.

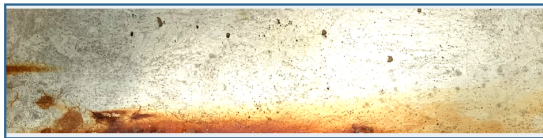
To keep augmented images within realistic appearance limits even when all three adjustments coincide at their extremes, the individual range was reduced from $[0.75, 1.28]$ to $[0.82, 1.18]$ —approximately $\pm 18\%$ per adjustment. In the worst case (all three parameters at the lower bound

Recipe 3 — Gaussian blur and sensor noise

blur radius [0.3, 1.5] px | noise sigma [0.005, 0.035] on [0, 1]

Evidence of mild focus and pixel-noise variability.

Original

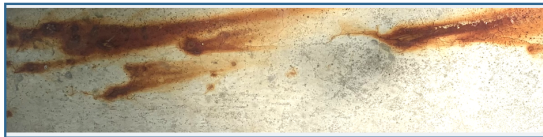


Class 1 | specimen F03 | F03-20240522-19W.png | mean Y=202.09

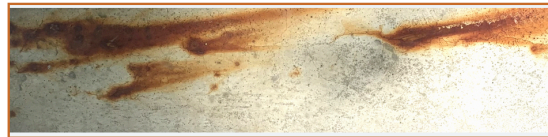
Augmented output



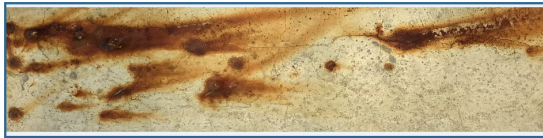
Parameters: blur_sigma=0.449349;noise_sigma=0.034666



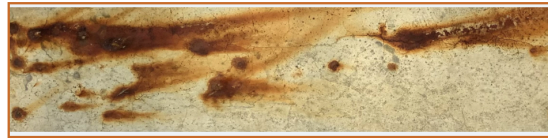
Class 2 | specimen D05 | D05-20240508-17W.png | mean Y=173.36



Parameters: blur_sigma=0.4487957;noise_sigma=0.024992



Class 3 | specimen D05 | D05-20240619-23W.png | mean Y=152.17



Parameters: blur_sigma=1.352402;noise_sigma=0.009466



Class 4 | specimen D04 | D04-20240508-17W.png | mean Y=113.03



Parameters: blur_sigma=0.374060;noise_sigma=0.033429

Examples are existing offline files from the generated dataset; no recipe was recomputed or changed for this figure.

Figure 5: Recipe 3 (slight blur and camera noise): one original per class (left) and its augmented copy (right). The effect is intentionally subtle at source resolution; rust patterns are fully preserved.

simultaneously), the compound effect reduces visual intensity by approximately $1 - 0.82^3 \approx 45\%$. In practice, three independent random values all landing near their extreme simultaneously is unlikely, and the typical combined effect is much smaller.

The range $[0.82, 1.18]$ is a conservative engineering choice. No separate measurement was made to determine this specific boundary. The mathematical argument for narrowing relative to single-parameter recipes is sound; the precise value of 0.82 is a judgement.

See: Figure ?? for visual examples.

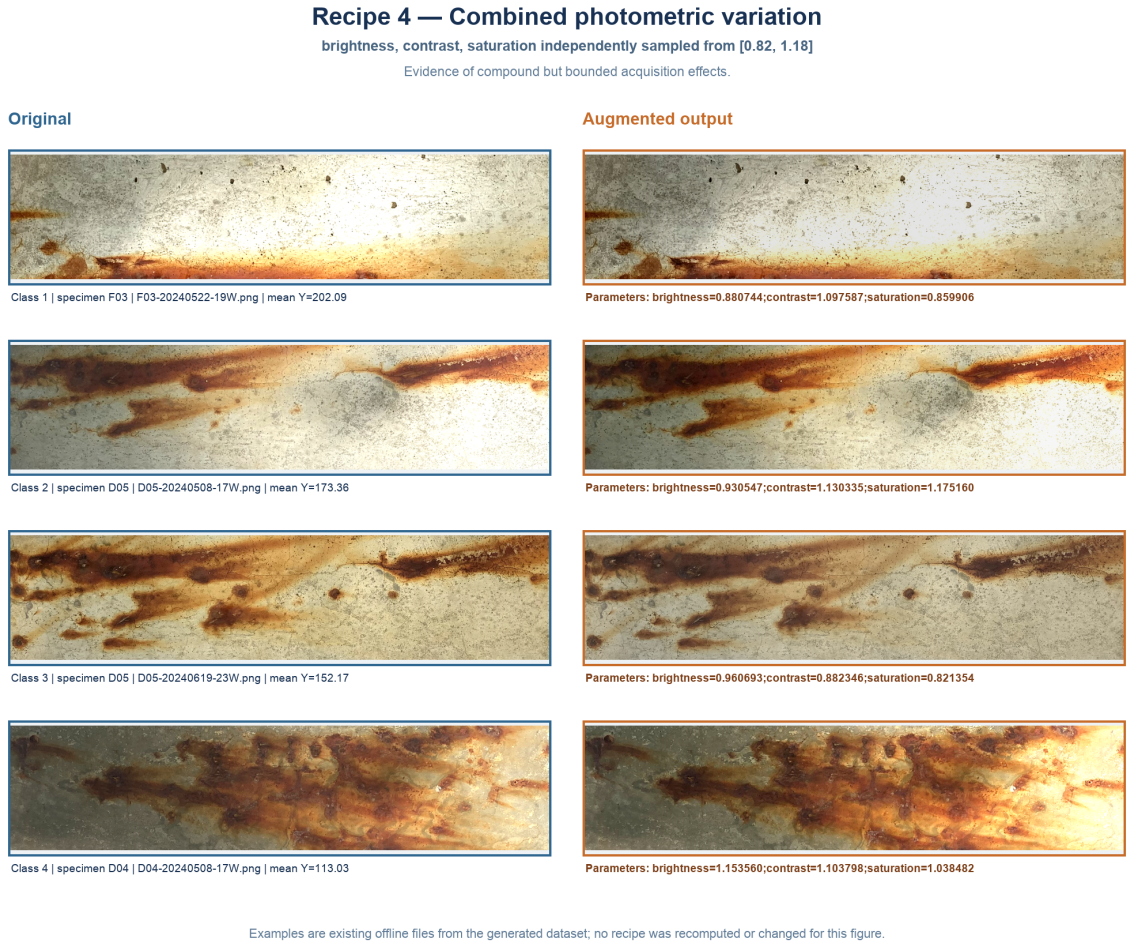


Figure 6: Recipe 4 (combined brightness, contrast, and colour intensity): one original per class (left) and its augmented copy (right). Simultaneous adjustment of three photometric properties simulates a photograph taken under broadly different conditions.

4.5 Recipe 5 — Left-Right Mirror and Brightness

What this recipe does

The image is reflected left-to-right (a horizontal mirror), and a mild brightness adjustment is applied to ensure the copy is not a purely deterministic duplicate of the original.

Why the mirror is safe for this classification label

The classification target, `A_Total_Rust_Category_(1-4)`, measures the total fraction of the specimen surface covered by rust. This is a scalar quantity—it answers the question “how much rust is present?” not “where on the specimen is the rust?” The total coverage does not change when the image is mirrored: a specimen showing 20% total rust coverage is still class 2 whether the rust patch appears on the left or the right.

This argument applies specifically to this label. It would *not* apply to the peak rust location column (`B_Location_of_Peak_Rust_in_length_[cm]`), which records a position. That column is preserved unchanged in the augmented CSV and must not be used as a spatial feature for mirrored copies.

Why vertical mirror and large rotations were excluded

A vertical mirror would show the specimen photographed upside-down. The 48 specimens were always photographed in the same fixed orientation; an upside-down image is not a physically possible acquisition condition for this setup.

Large rotations (more than approximately 10°) introduce empty corner regions that must be filled with an artificial value. Even with the best fill strategy (reflecting the edge pixels), these regions do not represent real specimen content and can be learned as spurious features. Rotations beyond 10° also do not correspond to plausible camera placement variability for a fixed imaging setup.

Brightness range [0.85, 1.15]

The brightness component of this recipe uses a range of [0.85, 1.15]. This is narrower than the [0.75, 1.28] range of Recipe 1, for two reasons: the mirror already provides strong spatial diversity, so the brightness component is secondary; and a narrower range avoids creating copies that are both mirrored and extremely dark. The specific value 0.85 is a conservative engineering choice with no separate measurement basis.

See: Figure ?? for visual examples.

5 Rejected Augmentation Techniques

Table ?? lists techniques that were considered and explains why each was excluded.

Table 9: Augmentation techniques considered and excluded.

Technique	Reason for exclusion
MixUp	Blends two images together and adjusts the class label proportionally. The result—for example, “60% class 2 and 40% class 3”—is not a physically meaningful rust severity. The four-class labels used in this study are discrete; they cannot be meaningfully interpolated.

Table ?? continued from previous page

Technique	Reason for exclusion
CutMix	Pastes a rectangular patch from one image into another and adjusts the label proportionally to the pasted area. A patch of clean concrete pasted into a severe-rust image creates a contradictory training example: the label suggests moderate severity, but a region of the image shows no rust at all. For specimens in classes 2 and 3, where total rust coverage is already limited, this creates the largest risk of label corruption.
Synthetic image generation (GAN, diffusion)	Generating new corrosion images using a generative model would require training a separate model on fewer than 20 images per class for classes 3 and 4. Models trained on so few examples produce unstable or repetitive outputs. This technique is not among the professor’s specified families and is outside the scope of this study.
Elastic deformation	Warpes the image as if the surface were made of elastic material. This technique was developed for medical imaging (deformable tissue in histology slides and MRI scans). A flat ferrocement slab photographed perpendicular to its surface does not deform in this way; applying elastic deformation produces physically implausible specimen images.
Random region removal (Cutout)	Replaces a random rectangular region of the image with black pixels. If the removed region contains the primary rust patch, the model receives an image labelled “moderate rust” that visually shows mostly clean concrete. For class 3 (20 images, limited rust coverage), this risk is highest.
Heavy colour shift ($> \pm 10\%$)	Shifting the hue of the image by more than approximately $10\text{--}18^\circ$ on the colour circle can move rust pixels from their orange-red range into yellow, green, or magenta—colours that do not occur in corroded ferrocement.
Vertical mirror	Produces an upside-down photograph of the specimen. Not a plausible acquisition condition for this fixed imaging setup.
Large rotation ($> \pm 10^\circ$)	Creates corner fill regions that are not representative of real specimen content. The imaging setup is fixed and does not produce rotated images.

6 Dataset Expansion Factor

Resulting dataset

Table ?? shows the dataset size before and after augmentation.

Literature basis for the $6\times$ factor

The expansion factor of $6\times$ (one original plus five augmented copies) was chosen to match the range supported by published corrosion studies of comparable dataset size. Table ?? lists the most relevant references.

The most directly comparable study [?] (~ 800 images, corrosion classification, offline augmentation) used $6.25\times$. The general survey [?] identifies $2\text{--}3\times$ as optimal for larger datasets

Recipe 5 — Horizontal flip and brightness

left-right mirror | brightness [0.85, 1.15]

Evidence of position-invariant spatial diversity for total-rust classification.

Original

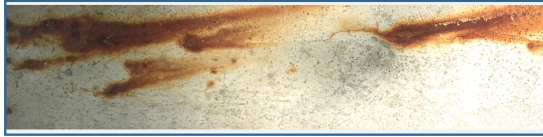
Augmented output



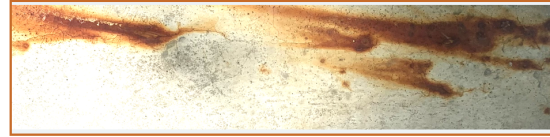
Class 1 | specimen F03 | F03-20240522-19W.png | mean Y=202.09



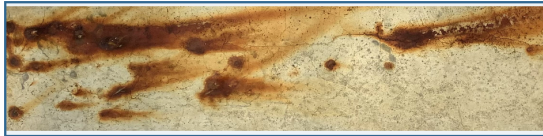
Parameters: brightness=1.016313;horizontal_flip=true



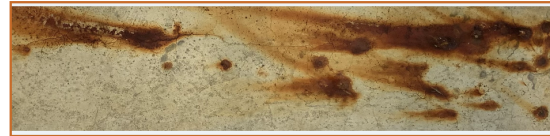
Class 2 | specimen D05 | D05-20240508-17W.png | mean Y=173.36



Parameters: brightness=1.115258;horizontal_flip=true



Class 3 | specimen D05 | D05-20240619-23W.png | mean Y=152.17



Parameters: brightness=0.851278;horizontal_flip=true



Class 4 | specimen D04 | D04-20240508-17W.png | mean Y=113.03



Parameters: brightness=1.149450;horizontal_flip=true

Examples are existing offline files from the generated dataset; no recipe was recomputed or changed for this figure.

Figure 7: Recipe 5 (left-right mirror and brightness): one original per class (left) and its mirrored-and-brightness-adjusted copy (right). The mirror effect is clearly visible; total rust coverage is unchanged.

Table 10: Dataset size before and after augmentation.

Class	Original images	Augmented copies	Total	Factor
1	658	3 290	3 948	6×
2	99	495	594	6×
3	20	100	120	6×
4	14	70	84	6×
Total	791	3 955	4 746	6×

Table 11: Expansion factors in comparable published studies.

Study	Architecture	Dataset size (pre-aug.)	Expansion	Result
Tank corrosion [?]	EfficientNetB0	~800 images	6.25 $\times$	94% accuracy
Steel corrosion [?]	CNN	100 images	20 $\times$	—
General survey [?]	Various	Various	2–3 $\times$ optimal	Diminishing returns above 3 $\times$
Steel defect detection	ResNet50, InceptionV3	800–2 400	3 $\times$	>90% accuracy

but acknowledges that corrosion-specific studies routinely use 3–6 $\times$ on smaller collections. The chosen 6 $\times$ sits at the upper boundary of this range.

Why 6 $\times$ was chosen over a higher factor

Using each of the five recipes exactly once per source image gives 6 $\times$ and makes the composition of the training set transparent: every unique augmentation that the pipeline can produce appears exactly once. Generating more copies would require repeating recipes with different random parameters, introducing diminishing diversity per additional copy. No published corrosion study of comparable size justifies an expansion beyond 6–10 $\times$, and the general survey [?] reports minimal accuracy improvement above 3 $\times$. Six copies from five distinct recipe families is the natural stopping point.

7 Reproducibility

How the same augmented images are generated every time

Every augmented image is produced using a random number that depends on three inputs: the global seed, the source image filename, and the copy number. These three inputs are combined using a standard cryptographic hash function (SHA-256), and the result is used to initialise the random number generator independently for each image-copy pair. Changing any one of the three inputs produces a completely different sequence of random numbers.

This means the output of the script is fully deterministic. Running the script again with the same global seed produces byte-identical images. Running it on a different computer or with different numbers of parallel processes produces the same result. The order in which images are processed does not affect the outcome.

Global seed

The global seed used for all augmentation is **20260630**. This value was chosen to match the date of the dataset generation run and has no mathematical significance beyond uniqueness.

Reproduction command

```
python augmentation/augment_dataset.py --copies 5 --seed 20260630 --
  overwrite
```

Provenance fields in the CSV

Every row in `Data/Images_Dataset_A-Z-1_augmented.csv` contains:

- `original_image_name`: the source file this row was derived from
- `augmentation_type`: the recipe name applied
- `augmentation_parameters`: the exact numerical values used (for example, `brightness=0.874061;horizo`)
- `is_augmented`: whether this row is an original or an augmented copy
- `specimen_id`: the specimen identifier, required for leakage-safe train/test splitting

8 Splitting: Augmented Images Belong to the Training Partition Only

Augmented images must not appear in the validation or test sets. The validation and test sets measure how well the model performs on images that look like real photographs. Augmented images do not look exactly like real photographs—they are intentionally altered. Including them in evaluation would measure performance on artificially varied images rather than on the real-world distribution.

More importantly, no augmented copy of a validation or test specimen may appear in the training set. Because augmented copies retain the visual identity of their source specimen (same rust pattern, same spatial layout, with only adjusted brightness or colour), placing a copy in training while the original is in the test set constitutes a data leak: the model has effectively seen the specimen before, in a slightly different form.

The split manifests produced by `augmentation/make_splits.py` implement these rules:

- **Training rows:** all rows (original + augmented) for the 38 training specimens.
- **Validation rows:** original rows only for 5 validation specimens.
- **Test rows:** original rows only for 5 test specimens.
- The 750 augmented copies of validation and test specimens are excluded from all splits.

9 Evidence Status Summary

Table ?? records the evidence basis and confidence level for every parameter choice in the pipeline.

10 Limitations

Contrast and colour intensity ranges are not measured from the dataset. The parameter ranges for light-to-dark contrast ($[0.78, 1.25]$) and colour intensity ($[0.78, 1.28]$) are drawn from published literature and set conservatively. A more rigorous calibration would measure the within-class variation in these properties and derive the range from that measurement, as was done for brightness. This analysis requires separating variation due to imaging conditions from variation due to physical specimen state (which correlates with the class label), and was not performed.

Blur is largely removed by downsampling. The blur applied at source resolution (2835×650 pixels) is reduced to 0.12–0.59 effective pixels at training resolution (224×224 pixels) by the

Table 12: Evidence basis for every parameter choice.

Parameter	Value	Evidence basis	Confidence
Brightness lower bound	0.75	Measured: $1/(199.1/144.8) = 0.727 \rightarrow 0.75$	High
Brightness upper bound	1.28	Measured: $253.1/199.1 = 1.271 \rightarrow 1.28$	High
Contrast lower bound	0.78	Literature: industrial inspection $\pm 15\text{--}25\%$	Medium
Contrast upper bound	1.25	Literature: industrial inspection $\pm 15\text{--}25\%$	Medium
Colour intensity lower	0.78	Conservative heuristic; literature $\pm 20\text{--}30\%$	Medium
Colour intensity upper	1.28	Conservative heuristic; literature $\pm 20\text{--}30\%$	Medium
Colour shift per channel	$\pm 6\%$ (0.94–1.06)	Measured: within-class R/G std $\approx 3\%$, range $\approx 11\%$; chosen range covers this conservatively	Medium–high
Blur radius at source	0.3–1.5 px	Literature: industrial inspection 0.1–2.0 px	Medium
<i>Blur at training res.</i>	<i>0.12–0.59 px</i>	<i>Derived from resize scale 0.394 (disclosure)</i>	<i>High</i>
Noise level at source	0.005–0.035	Literature: surface inspection 0.01–0.05	Medium
<i>Noise at training res.</i>	<i>$\sim 0.003\text{--}0.022$</i>	<i>Derived from resize scale (disclosure)</i>	<i>High</i>
Combined recipe range	0.82–1.18	Engineering heuristic: $0.82^3 = 0.551$, keeps compound accumulation within $\approx 50\%$	Medium
Mirror flip	Fixed (no range)	Logical: total coverage is position-invariant	High
Flip brightness	0.85–1.15	Conservative sub-range of Recipe 1; heuristic	Low–medium
Expansion factor	$6\times$	Literature: PMC11829104 ($6.25\times$); survey (2–3 $\times$ optimal for larger sets)	High

resize operation during model training. The blurring contribution to model regularisation is therefore smaller than the stated source-resolution values suggest. Noise is similarly attenuated (to approximately 5.6 intensity units maximum at training resolution). Both effects are real and contribute to regularisation, but quantitatively less than the stated source-resolution values.

The class imbalance is preserved. Augmentation multiplies all classes by the same factor ($6\times$). Class 1 remains 83.2% of the dataset. The model training must use class-weighted loss or an equivalent technique to address this imbalance.

The augmented dataset is fixed. Generating images offline means that only five specific random realisations of each recipe appear per source image. During model training, no new random variations are created per epoch. Online augmentation (applying transformations dynamically at training time) would produce greater variety at the cost of less convenient inspection and archiving. The offline approach was chosen for transparency and reproducibility.

11 Figures and Data Files

Table ?? lists all figure and data files produced by the augmentation pipeline.

Table 13: Figures and data files produced by the augmentation pipeline.

File	Content
augmentation/figures/luminance_histogram.png	Distribution of overall image brightness across all 791 images, with p_5 (144.8), median (197.7), and p_{95} (253.1) marked. Shows the $1.75\times$ spread that motivates the brightness range.
augmentation/figures/luminance_extremes.png	Side-by-side comparison of the four darkest images (specimen D04, class 4, brightness 98–113) and four brightest images (G-series, class 1, brightness 255). Shows the full $2.60\times$ real-world variation.
augmentation/figures/recipe1_brightness_contrast.png	One original image per class and its brightness-contrast augmented copy.
augmentation/figures/recipe2_saturation_colour_balance.png	One original image per class and its colour-intensity-and-shift augmented copy.
augmentation/figures/recipe3_blur_noise.png	One original per class and its blur-and-noise augmented copy. Effect is intentionally subtle at source resolution.
augmentation/figures/recipe4_combined.png	One original per class and its combined-adjustment augmented copy.
augmentation/figures/recipe5_flip_brightness.png	One original per class and its mirrored-and-brightness-adjusted copy.
augmentation/tables/image_statistics_full.csv	Per-image statistics for all 791 source images: brightness, per-channel means, channel standard deviations, file metadata, class label, specimen ID (791 rows).
Data/Images_Dataset_A-Z-1_augmented.csv	4746-row provenance table: one row per output image, with source filename, recipe, exact parameters, label, and specimen ID.

Python library versions

```
Pillow    -- image loading, colour adjustment, blur, flip  
NumPy     -- noise generation, array arithmetic  
openpyxl  -- Excel workbook reading
```

Exact installed versions are recorded in `augmentation/requirements.txt`.

References

- [1] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, “Training data-efficient image transformers & distillation through attention,” in *Proceedings of the 38th International Conference on Machine Learning (ICML 2021)*, vol. 139, pp. 10347–10357, 2021.
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- [4] C. Shorten and T. M. Khoshgoftaar, “A survey on image data augmentation for deep learning,” *Journal of Big Data*, vol. 6, no. 1, p. 60, 2019. [doi:10.1186/s40537-019-0197-0](https://doi.org/10.1186/s40537-019-0197-0) (arXiv:1906.11887).